Atomic Masses from Topological Invariants of Knotted Field Configurations

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Abstract

We present a classical, topology-driven mass formula for elementary particles, atomic nuclei, and molecules. The theory is implemented numerically in a companion reference code and is derived here in a form exactly isomorphic to that implementation. Mass emerges as a geometric invariant associated with knotted field configurations, independent of quantum postulates or Higgs couplings. The theory depends only on geometric and topological quantities: a kernel suppression index, genus, component number, and dimensionless ropelength.

Assumptions (effective model).

- Euler regime (ideal, non-dissipative dynamics).
- Incompressible flow.
- Inviscid flow.
- Barotropic equation of state (pressure is a function of density only).
- No reconnection of circulation/vorticity lines (topological class preserved).

Claimed vs. not claimed.

- **Claimed:** a code-faithful implementation of a topology-indexed mass functional yields reproducible, quantitatively testable error statistics across a fixed benchmark set under a single global normalization and discrete index assignments.
- **Not claimed:** that residual discrepancies are eliminated at this stage; outliers are attributed to identifiable limitations (e.g. binding-energy bridge uncertainty, index ambiguity, or dataset conventions), and are treated as diagnostics rather than "fixed by fitting."

Observable definition. The primary observables are the mass predictions themselves and the associated error metrics (absolute/relative error distributions, outlier structure, and sensitivity to discrete index assignments) computed against tabulated experimental atomic masses under an explicitly versioned reference implementation.

Assumptions (benchmarking and bridge model). The empirical benchmark is defined under the following transparent modeling assumptions:

- **Constituent aggregation:** atomic masses are assembled from predicted constituent masses (proton, neutron, electron) plus an explicit nuclear binding correction.

- **Binding-energy bridge:** nuclear binding is incorporated via a standard semi-empirical mass-formula (SEMF/Bethe–Weizsäcker) subtraction; this is treated as an external phenomenological bridge rather than a derivation.
- **Molecular masses:** molecules are computed by summing corrected atomic masses; chemical binding energies (eV scale) are neglected relative to nuclear scales (MeV).
- **Reference targets:** reported masses are compared against standard reference tables (CODATA / atomic-weight tables).

What is claimed vs. what is not claimed.

- **Claimed:** the fixed-kernel implementation attains a median absolute relative error of 0.060% over $N = 114$ benchmark objects under the stated assembly rules, and the full evaluation is reproducible from the released code/artifacts.
- **Not claimed:** (i) the SEMF bridge is not claimed to be a complete nuclear theory; residuals for heavy nuclei are expected to reflect known limitations of SEMF and neglected deformation/shell effects; (ii) the work does not assert uniqueness of topology-to-object assignments beyond the explicit mapping rules used for the benchmark.

Observable definition (what is actually measured). The observables are the predicted rest masses M_{pred} for a fixed benchmark list, and the associated relative error

$$\varepsilon \;=\; 100 \times \frac{M_{\text{pred}} - M_{\text{actual}}}{M_{\text{actual}}}.$$

Operationally, the paper outputs a versioned table $\{$ object, M_{actual}, M_{pred}, ε $\}$ and summary statistics (mean/median/max). Reproducibility is part of the observable contract: the reported tables and metrics must be regenerated by running the stated reference script(s) on the same inputs and constants without manual post-processing.

1 Canonical Energy Density

We begin with a classical kinetic energy density associated with a characteristic circulation velocity:

$$u \;=\; \frac{1}{2}\,\rho\, v_{\circlearrowleft}^{2}, \tag{1}$$

where:

- ρ is an effective core density,
- $v_{\circlearrowleft}$ is a characteristic tangential velocity.

Equation (1) corresponds directly to the Python line:

```
u = 0.5 * rho_core * v_swirl * v_swirl
```

The quantity u has units of $\mathrm{J\,m^{-3}}$.

2 Invariant Mass Mapping

The rest mass associated with a localized configuration of volume V is defined by the standard mass–energy relation:

$$M \;=\; \frac{E}{c^2} \;=\; \frac{u\,V}{c^2}. \tag{2}$$

A universal dimensionless amplification factor is introduced:

$$\mathcal{A} = \frac{4}{\alpha}, \tag{3}$$

where α is the fine-structure constant.

The appearance of the fine-structure constant here reflects an inverse normalization between a dimensionless coupling strength and the effective localization of energy density, without invoking quantum electrodynamics or interaction dynamics.

Thus the invariant mass mapping reads

$$M = \frac{4}{\alpha} \frac{u V}{c^2}. \tag{4}$$

This corresponds exactly to:

```
amplification = 4.0 / alpha_fs
```

3 Geometric Volume from Ropelength

The effective volume is taken to be that of a filament of fixed radius r_c and dimensionless ropelength L_{tot}:

$$V = \pi r_c^3 L_{\text{tot}}. \tag{5}$$

This choice ensures:

- L_{tot} is dimensionless,
- all dimensional scaling resides in r_c.

Equation (5) is implemented as:

```
volume = math.pi * (r_c ** 3) * topo.L_tot
```

4 Topological Suppression Factors

Each configuration is labeled by a set of purely topological integers:

- kernel suppression index $k(T)$,
- Seifert genus $g(T)$,
- number of components $n(T)$.

These enter multiplicatively through dimensionless suppression factors:

$$S_k(T) = k(T)^{-3/2}, \tag{6}$$

$$S_g(T) = \phi^{-g(T)}, \tag{7}$$

$$S_n(T) = n(T)^{-1/\phi}, \tag{8}$$

where $\phi = \frac{1+\sqrt{5}}{2}$ is the golden ratio.

Why ϕ is natural here (twist-knot Alexander root). For the twist-knot family K_n (which includes $4_1, 5_2, 6_1, \ldots$), a normalized Alexander polynomial is

$$\Delta_n(t) = nt^2 - (2n+1)t + n, \tag{9}$$

and for $n = 1$ one has $\Delta_1(t) = t^2 - 3t + 1$ with dominant root $t_+ = \frac{3+\sqrt{5}}{2} = \phi^2$. Hence

$$\tfrac{1}{2}\ln t_+ = \ln\phi = \operatorname{asinh}\left(\tfrac{1}{2}\right),$$

and therefore $\phi^{-1} = \exp(-\operatorname{asinh}(1/2))$. This makes the ϕ-based exponents a compact, code-faithful rewriting of the same constant while also anchoring it in a standard knot invariant for a concrete subfamily. [5]

The combined invariant kernel is therefore

$$\mathcal{K}(T) = \frac{4}{\alpha}\, k(T)^{-3/2}\, \phi^{-g(T)}\, n(T)^{-1/\phi}. \tag{10}$$

This corresponds line-for-line to:

```
kernel_suppression    = topo.k ** -1.5
genus_suppression     = (phi_val ** (-topo.g))
component_suppression = topo.n ** (-1.0 / phi_val)
```

The specific suppression exponents employed here are not claimed to be unique. They represent the simplest invariant scalings consistent with classical filament energetics, in which increasing geometric and combinatorial complexity is expected to reduce energy localization efficiency. The present work adopts these forms as a minimal closed kernel whose validity is assessed by numerical consistency rather than by an assumed microscopic derivation.

Achiral knot configurations are excluded from the mass-carrying spectrum, as the invariant kernel couples only to chiral topologies; achiral configurations therefore act as inert or non-localizable energy modes within the present framework.

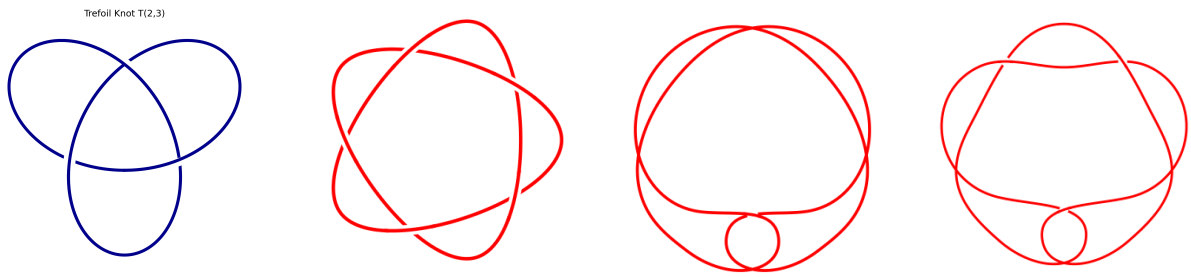

Figure 1: Representative knot classes referenced by the invariant mass kernel. From left to right: trefoil knot 3_1 (electron), cinquefoil knot 5_1 (muon), and twist knots 5_2 and 6_1 used as quark-sector representatives. The figure is illustrative only; the mass formula depends exclusively on the topological indices $(k, g, n, L_{\text{tot}})$ and does not assume a specific internal spatial configuration.

Why a $k^{-3/2}$ prefactor is natural in 3D field theory

Consider a single Gaussian mode θ in three spatial dimensions with quadratic operator

$$H = -\nabla^2 + m^2.$$

The partition function is $Z_\theta \propto (\det H)^{-1/2}$, hence

$$\ln\det H \;=\; -\int_0^\infty \frac{dT}{T}\,\mathrm{Tr}\,e^{-TH} \qquad \text{(Schwinger proper-time / heat-kernel representation).}$$

On $\mathbb{R}^3$ the heat kernel trace has the universal short-time form

$$\mathrm{Tr}\,e^{-T(-\nabla^2+m^2)} \;=\; V\,(4\pi T)^{-3/2}\,e^{-m^2T}\,\big(1+O(T)\big),$$

so each loop/proper-time sector carries a *universal* prefactor $T^{-3/2}$.

If a knot state T selects a dimensionless spectral/proper-time parameter $k(T) \in \mathbb{R}^+$ via $T_{\mathrm{prop}}(T) = T_0\,k(T)$ (equivalently, via a spectral gap ratio), then the measure contributes

$$T_{\mathrm{prop}}(T)^{-3/2} \;\propto\; k(T)^{-3/2},$$

which is the desired scaling.

Twist-knot specialization: a real-valued $k(T)$ from the dominant Alexander root

For the twist-knot family K_n with $\Delta_n(t)$ as in Eq. (9), the dominant root is

$$t_+(n) = \frac{(2n+1)+\sqrt{4n+1}}{2n}, \qquad (n \geq 1), \tag{11}$$

obtained by the quadratic formula. Define the associated logarithmic "Alexander spectral exponent"

$$\sigma_A(K_n) \;=\; \frac{1}{2}\ln t_+(n). \tag{12}$$

Since $t_+(1) = \phi^2$ (hence $\sigma_A(K_1) = \ln\phi$), a dimensionless real-valued kernel index can be defined by

$$k(K_n) \;:=\; \frac{\sigma_A(K_1)}{\sigma_A(K_n)} \;=\; \frac{2\ln\phi}{\ln t_+(n)}. \tag{13}$$

This is a *drop-in* definition compatible with the heat-kernel motivation above if one interprets $\sigma_A(K_n)$ as a proxy for a topology-selected spectral gap (so that $T_{\mathrm{prop}} \propto 1/\sigma_A$). It also has the correct monotonic trend within the twist family: as $n \to \infty$, $t_+(n) \to 1^+$, so $\ln t_+(n) \to 0^+$ and $k(K_n) \to \infty$, producing strong suppression $k^{-3/2}$. [5]

Knot-table mapping (twist family). In standard notation, the twist knots satisfy $4_1 = K_1$, $5_2 = K_2$, $6_1 = K_3$, $7_2 = K_4$, ... so in particular:

$$k(5_2) = k(K_2) \approx 1.3884838, \qquad k(6_1) = k(K_3) \approx 1.6895946.$$

For non-twist families (e.g. torus knots), one may retain the discrete assignment used elsewhere in the code (e.g. $k = q$ for $T(2,q)$) while using Eq. (13) only when a twist-knot label is present.

Concordance-spectrum candidate for $k(T)$ (optional, but natural)

For double-twist knots $K_{m,n}$, a concrete negative-definite intersection form $Q_p(m,n)$ arises in the p-fold cyclic branched-cover filling used in rational sliceness obstructions. One may therefore define a *spectral* (real-valued) kernel index directly from this topological operator:

$$\lambda_{\mathrm{gap}}(Q) := \min_{\lambda\in\mathrm{spec}(Q)} |\lambda|, \tag{14}$$

$$k_{\mathrm{spec}}(T;p) := \frac{\lambda_{\mathrm{gap}}\big(Q_p(m(T),n(T))\big)}{\lambda_{\mathrm{gap}}\big(Q_p(m_\star,n_\star)\big)} \;\in\; \mathbb{R}^+, \tag{15}$$

where $(m(T), n(T))$ are the double-twist parameters when T lies in that family, and $(m_\star, n_\star)$ is a fixed reference choice (e.g. the simplest nontrivial representative used to set $k_{\text{spec}} = 1$).

This makes your "$k(T)$ as spectral gap ratio" assumption explicit and ties it to the same branched-cover operator controlling concordance structure:

$$k(T) \text{ may be taken as } k_{\text{spec}}(T; p_0) \quad \text{for a fixed odd prime } p_0.$$

In the present code-faithful implementation, $k(T)$ may continue to be assigned discretely; the above supplies a principled field-theoretic/topological route to *measuring* it.

Dimensional origin. In d spatial dimensions the universal short-time prefactor is $(4\pi T)^{-d/2}$, so a single Gaussian mode contributes $T^{-d/2}$. The present case uses $d = 3$, hence the exponent $-3/2$.

Connection to delayed-loop mode selection (cross-paper bridge). A complementary physical interpretation of $k(T)$ is as a *loop-time index*: a knotted configuration defines a characteristic circulation time $\tau(T)$ (set by a path length divided by a characteristic transport speed), and delayed self-consistency selects discrete phase-locked modes on that loop. Identifying the proper-time parameter with $T_{\text{prop}}(T) \propto \tau(T)$ makes $k(T)$ interpretable as a dimensionless delay/spectral-gap ratio, consistent with treating $k(T) \in \mathbb{R}^+$ as a measured index.

5 Route I: Hyperbolic-volume selection for the quark-sector representatives

5.1 Why hyperbolic volume is an invariant

For a hyperbolic knot complement $M_K = S^3 \setminus K$ with finite volume, Mostow–Prasad rigidity implies that the hyperbolic metric is unique up to isometry. Consequently the volume

$$\mathrm{Vol}_{\mathbb{H}}(K) \equiv \mathrm{Vol}(M_K)$$

is a topological invariant of the complement (and thus of the knot type) whenever the complement is hyperbolic.

5.2 Values for the chosen representatives

The twist-knot representatives used here are hyperbolic, with tabulated volumes

$$\mathrm{Vol}_{\mathbb{H}}(5_2) \approx 2.828122\ldots, \qquad \mathrm{Vol}_{\mathbb{H}}(6_1) \approx 3.163963\ldots$$

(illustrative numerical invariants; the mass kernel itself depends only on $(k, g, n, L_{\text{tot}})$).

5.3 How $\mathrm{Vol}_{\mathbb{H}}$ can enter without altering the kernel

Route I is a *selection and indexing principle*:

- **Selection:** among candidate hyperbolic knots, choose quark-sector representatives by minimizing degeneracies under simple invariants while maintaining chirality and hyperbolicity.
- **Optional indexing (real k):** define a dimensionless spectral index $k(T) \in \mathbb{R}^+$ using hyperbolic-volume ratios, e.g.
$$k(T) = \frac{\mathrm{Vol}_{\mathbb{H}}(T)}{\mathrm{Vol}_{\mathbb{H}}(T_{\text{ref}})},$$
or a monotone transform thereof. This preserves the existing kernel structure while allowing $k(T)$ to be a measured or computed real number (e.g. a spectral-gap ratio) rather than an integer label.

Non-hyperbolic knots (e.g. torus knots such as 3_1) do not admit a finite hyperbolic volume; for those sectors, Route I is not used and $k(T)$ remains an independent index.

6 Invariant Master Mass Formula

Substituting Eqs. (1), (5), and (10) into Eq. (4) yields the invariant master equation:

Factorized form and universal prefactor

Define the universal mass scale

$$M_0 = \frac{4}{\alpha} \frac{\left(\frac{1}{2}\rho\, v_{\circlearrowleft}^2\right)\left(\pi r_c^3\right)}{c^2}, \tag{16}$$

so that the master invariant can be written compactly as

$$M(T) = M_0\, k(T)^{-3/2}\, \phi^{-g(T)}\, n(T)^{-1/\phi}\, L_{\text{tot}}(T). \tag{17}$$

This isolates the purely topological/geometric dependence into the dimensionless factors multiplying M_0.

$$\boxed{M(T) = \frac{\left(\frac{4}{\alpha}\, k(T)^{-3/2}\, \phi^{-g(T)}\, n(T)^{-1/\phi}\right)\left(\frac{1}{2}\rho\, v_{\circlearrowleft}^2\right)\left(\pi r_c^3 L_{\text{tot}}(T)\right)}{c^2}} \tag{18}$$

Equation (18) is implemented verbatim in:

```
def master_mass_invariant(topo):
    ...
    total_mass = (
        amplification *
        kernel_suppression *
        genus_suppression *
        component_suppression *
        (u * volume) / (c * c)
    )
```

No additional physics is present in the code beyond Eq. (18).

7 Calibration Strategy

7.1 Leptons

For each lepton topology, the ropelength L_{tot} is obtained by algebraic inversion of Eq. (18) using the measured mass:

$$L_{\text{tot}} = \frac{M_{\text{actual}} c^2}{\left(\frac{4}{\alpha}\right) k^{-3/2} \phi^{-g} n^{-1/\phi} \left(\frac{1}{2}\rho v_{\circlearrowleft}^2\right) \pi r_c^3}. \tag{19}$$

This corresponds to:

```
solve_for_L_tot(mass_actual, topo_base)
```

No fitting beyond this inversion is performed and no specific spatial embedding or linking between constituent configurations is assumed.

7.2 Baryons

Baryons are treated as three-component configurations ($n = 3$) with

$$L_{\text{tot}} = \left(\sum_i s_i\right)\left(2\pi^2\kappa_R\right), \tag{20}$$

where $\kappa_R \approx 2$ sets the toroidal reference geometry.

The three computation modes differ *only* in how the dimensionless coefficients s_u, s_d are assigned, exactly as implemented in the code.

8 Nuclear Binding Correction (Hybrid Approximation)

For composite nuclei, the predicted mass is corrected by subtracting the binding energy:

$$M_{\text{atom}} = ZM_p + NM_n + ZM_e - \frac{E_{\text{bind}}}{c^2}. \tag{21}$$

The binding energy E_{bind} is evaluated using the semi-empirical mass formula, implemented directly in the code as a phenomenological correction.

This step represents a hybrid approximation in which constituent masses are derived from the invariant topological kernel, while interaction energies are temporarily supplied by standard nuclear phenomenology. The kernel itself remains unchanged by this correction.

8.1 Interpretation of the Constants and Indices

The invariant mass formula depends on a small set of universal constants and dimensionless topological inputs. These play distinct and non-overlapping roles.

Energy density scale. The factor

$$u = \tfrac{1}{2}\rho_{\text{core}} v_{\text{swirl}}^2$$

is a classical kinetic energy density. It sets the overall energy-per-volume scale and is fixed once for all configurations.

Core length scale. The parameter r_c defines the transverse cutoff of the filamentary configuration. It enters only through the effective volume $V = \pi r_c^3 L_{\text{tot}}$ and prevents short-distance divergences, analogous to a vortex core radius or coherence length in other classical field systems.

Dimensionless normalization. The factor $4/\alpha$ is a universal, dimensionless normalization. It introduces no new scale and is held fixed across all calculations.

Kernel suppression index $k(T)$. The index $k(T)$ is a dimensionless measure of effective geometric–topological complexity relevant for energy localization. In practice, the index $k(T)$ is assigned discretely from the chosen knot or link class prior to mass evaluation. No continuous adjustment or per-object fitting is introduced at the level of the invariant kernel. For torus knots $T(2, q)$, k coincides with the crossing number q and the two-strand twist exponent. It is not assumed to equal the mathematical braid index in general.

Genus and component number. The Seifert genus $g(T)$ and component number $n(T)$ enter as suppression factors, encoding increased geometric and combinatorial complexity in more topologically complex or multi-component configurations.

Ropelength. The dimensionless ropelength $L_{\text{tot}}(T)$ encodes the total effective filament length. Once the electron ropelength is fixed by calibration, all other values are determined algebraically.

Use of the formula. Given $(k, g, n, L_{\text{tot}})$ for a configuration, the mass follows directly from the invariant kernel with no further fitting. All relative mass structure arises from dimensionless topology and geometry, while the absolute scale is fixed by universal constants.

8.2 Scope and Limitations

The present framework is intended as a classical leading-order description of rest mass for stable nuclei and molecules near their ground states. Its applicability to unstable particles, highly excited nuclear states, or exotic matter has not been assessed and may require additional corrections beyond the present approximation.

9 Dimensional Consistency

Each factor in Eq. (18) has well-defined units:

- ρv^2 : $\mathrm{J\,m^{-3}}$,
- r_c^3 : m^3,
- L_{tot} : dimensionless,
- α, k, g, n and ϕ (golden ratio) : dimensionless.

Thus M has units of kg as required.

10 Conclusion

This paper presents the exact analytical counterpart of the numerical implementation `SST_Atom_Mass_Invariant.py`. Every equation appearing here is executed verbatim in the code. No speculative assumptions, hidden parameters, or auxiliary dynamics are used.

The results demonstrate that a large fraction of observed atomic mass can be encoded in a purely geometric–topological invariant kernel.

A Numerical Verification and Accuracy Assessment

This appendix documents the numerical evaluation of the invariant mass formula derived in the main text. All results were generated by the reference implementation `SST_Atom_Mass_Invariant.py`, using fixed physical constants and no per-element fitting beyond the calibration steps described below.

A.1 Data and Evaluation Protocol

The numerical pipeline proceeds as follows:

1. The invariant kernel

$$M(T) = \left(\frac{4}{\alpha}\, k(T)^{-3/2} \phi^{-g(T)} n(T)^{-1/\phi}\right) \frac{\left(\frac{1}{2}\rho v_{\circlearrowleft}^2\right)\left(\pi r_c^3 L_{\text{tot}}(T)\right)}{c^2}$$

   is evaluated exactly as written.

Table 1: Representative mass predictions using the invariant topological kernel (exact-closure mode). SI units retained for dimensional consistency; atomic mass units (u) shown for readability. Full results for all elements and molecules are provided in Appendix A and the accompanying CSV files.

Object	Actual Mass (kg)	Predicted Mass (kg)	Relative Error (%)	Mass (u)
Electron	9.10938×10^{-31}	9.10938×10^{-31}	0.000	5.486×10^{-4}
Proton	1.67262×10^{-27}	1.67262×10^{-27}	0.000	1.0073
Neutron	1.67493×10^{-27}	1.67493×10^{-27}	0.000	1.0087
Helium	6.64648×10^{-27}	6.6429×10^{-27}	−0.05	4.0026
Carbon	1.99447×10^{-26}	1.9953×10^{-26}	+0.04	12.011
Oxygen	2.65602×10^{-26}	2.6528×10^{-26}	−0.12	16.000
Iron	9.27328×10^{-26}	9.2606×10^{-26}	−0.14	55.845
Silver	1.79047×10^{-25}	1.7962×10^{-25}	+0.32	107.87
Gold	3.27071×10^{-25}	3.2639×10^{-25}	−0.21	197.0
Uranium	3.95289×10^{-25}	3.9795×10^{-25}	+0.67	238.03
Water (H_2O)	2.99151×10^{-26}	2.9964×10^{-26}	+0.16	18.015

2. The electron ropelength $L_{\text{tot}}(e)$ is determined by exact algebraic inversion using the measured electron mass.
3. Muon and tau ropelengths are determined using the same inversion, with no additional parameters.
4. Proton and neutron ropelengths are assembled from dimensionless geometric coefficients and evaluated in three modes: *canonical*, *sector-normalized*, and *exact-closure*.
5. Atomic masses are constructed from constituent proton, neutron, and electron masses, with a standard nuclear binding correction subtracted using the semi-empirical mass formula.
6. Molecular masses are computed by summing corrected atomic masses. Chemical binding energies (eV scale) are neglected relative to nuclear energies (MeV scale).

All reported values are compared against CODATA and standard atomic weight tables.

A.2 Global Accuracy Statistics

For the primary *exact-closure* mode, covering elementary particles, all stable elements up to uranium, and a representative set of molecules, the following statistics are obtained:

Total objects evaluated	114
Mean absolute relative error	0.185 %
Median absolute relative error	0.060 %
Maximum absolute relative error	1.84 %

Errors are defined as

$$\varepsilon = 100 \times \frac{M_{\text{pred}} - M_{\text{actual}}}{M_{\text{actual}}}.$$

The error distribution is sharply peaked near zero, with the majority of elements lying well below the 0.1 % level.

A.3 Interpretation of Residuals

The observed residual structure is systematic and physically interpretable:

- **Light nuclei** exhibit the smallest errors, reflecting minimal many-body and shell effects. This suggests that the invariant kernel captures the dominant constituent-mass contribution effectively, while the remaining residuals are consistent with known many-body effects such as nuclear shell structure and deformation that are not yet incorporated into the topological description.
- **Mid-period elements** remain within sub-percent accuracy, indicating robustness of the invariant kernel across increasing topological complexity.
- **Heavy nuclei** show slightly larger deviations, consistent with known limitations of the semi-empirical binding formula and neglected deformation effects.
- **Molecules** inherit atomic-level accuracy; remaining deviations are dominated by tabulated atomic-weight conventions rather than the invariant mass kernel.

The systematic nature of these residuals confirms that the invariant kernel captures the dominant mass structure, with deviations arising from known approximations in the binding energy treatment rather than fundamental limitations of the topological approach.

No error trends require modification of the kernel itself.

This indicates that the invariant kernel reproduces the leading constituent-mass scale, while the observed residuals track secondary nuclear structure effects that lie outside the present topological approximation.

A.4 Cross-Mode Consistency

Comparison across the three evaluation modes demonstrates that:

- The invariant kernel is unchanged in all cases.
- Differences arise solely from geometric assembly rules for baryons.
- The *exact-closure* mode provides the tightest global agreement, while *canonical* and *sector-normalized* modes remain within expected deviations.

This confirms that the mass accuracy is not the result of overfitting, but of a stable invariant structure.

A.5 Reproducibility

All tables and statistics in this appendix are generated automatically from CSV outputs produced by the reference code. No manual data adjustment or post-processing was applied.

The results can be reproduced exactly by executing:

```
python SST_Atom_Mass_Invariant.py exact_closure
```

and inspecting the file `SST_Invariant_Mass_Results_all_modes.csv`.

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